TechCampus

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Clinical Simulation: An Interactive Simulation Platform for Clinical Reasoning in Emergency Medicine

Course: Educational Technologies (SoSe 2026)

Presented by: Julian Krauß, Florian Hauptmann, Robin Hefner

Clinical Simulation: Project Scope & Team Roles

- **Project Goal:** Interactive web simulation to train medical students in clinical reasoning and cost-effective decision-making.
- **Core Vision:** Shift learning from passive factual memorization to active, inquiry-based clinical problem-solving.
- **Tech Design:** Zero-install, highly responsive, and accessible on any device.

Team Responsibility

R. Hefner Visual layout, responsive UI/UX, and clinical case writing.

F. Hauptmann Core simulation mechanics, emergency room workflow, and difficulty engine.

J. Krauß Dual-axis assessment, analytics dashboard, and the 10-trial block reflection system.

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Why Do Medical Students Need Simulation Environments?

The "Clinical Reasoning" Gap

Traditional lectures focus on passive facts. Students struggle to apply knowledge under real ER pressure.

Cognitive Overload

Novices are overwhelmed by diagnostic options, leading to costly "shotgun-testing" patterns.

The Safe Sandbox

A risk-free environment where students experience economic consequences without risking patient safety.

The Solution

A realistic ER simulation that scales difficulty and provides structured feedback for deep reflection.

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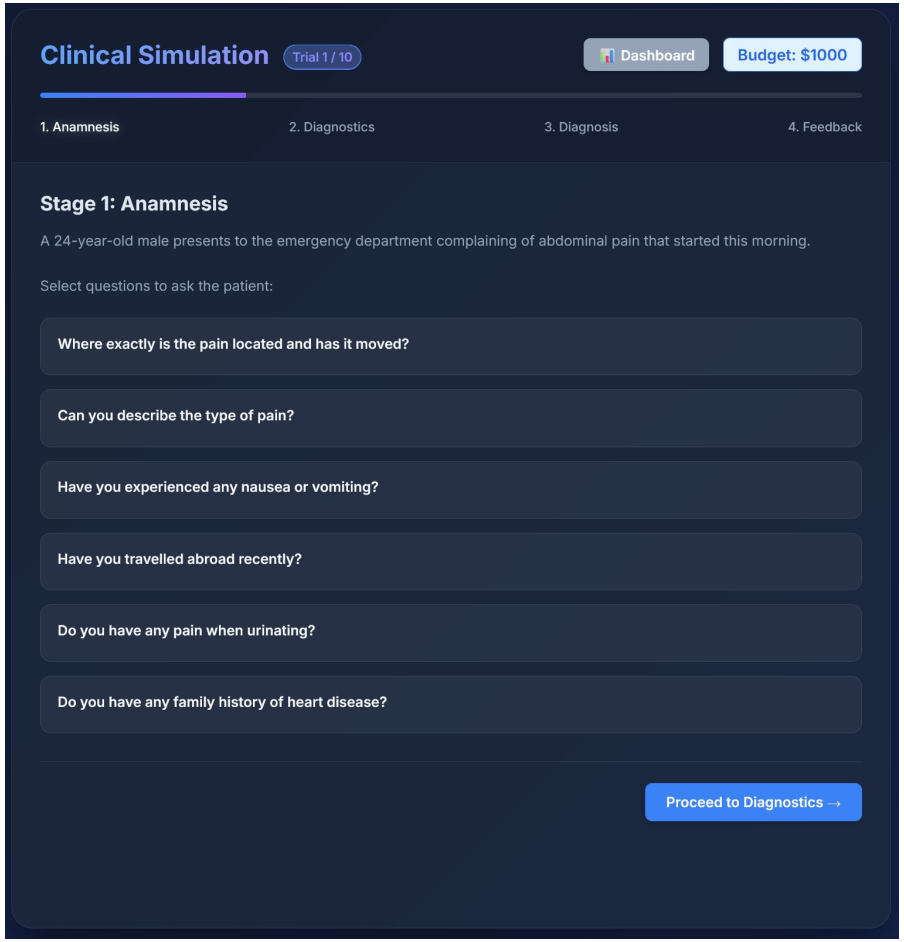
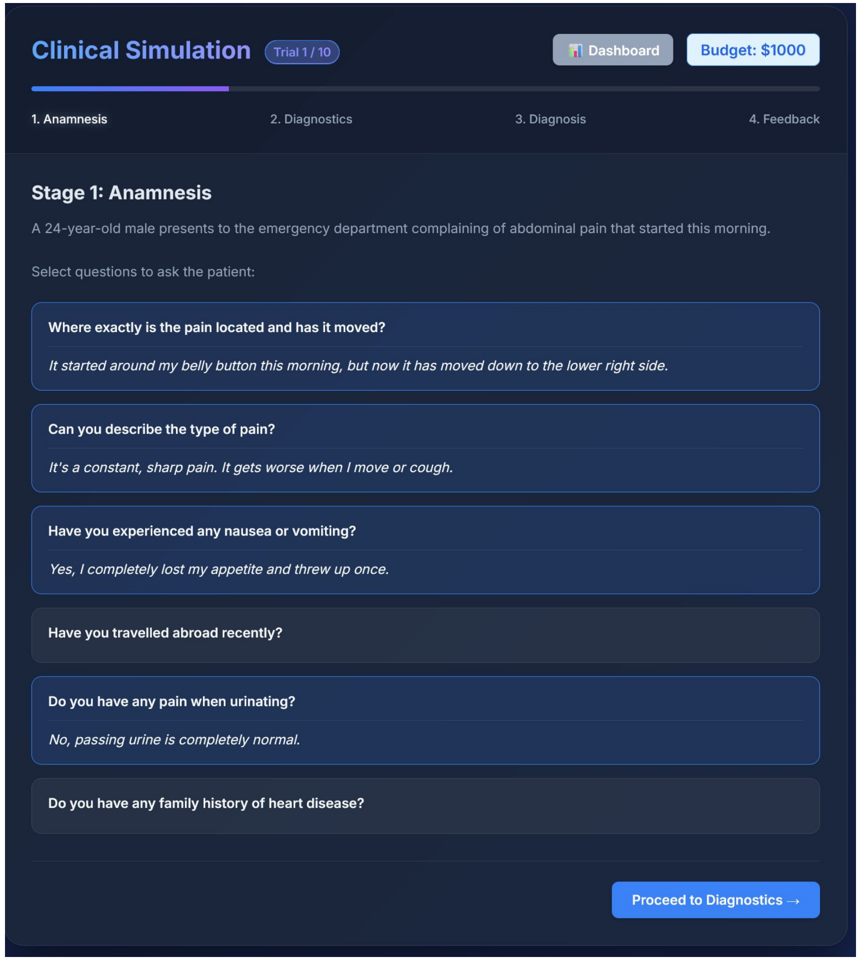
PBL in Action: Authentic, Ill-Structured Cases

- **Problem-Based Learning (Barrows, 1986):** Learning is anchored in complex, open-ended clinical problems mirroring real-world practice.
- **Chief Complaint:** The student starts with only basic intake details (e.g., *"A 24-year-old male presents to the emergency department …"*).
- **Self-Directed Inquiry:** No initial diagnostic guidance or multiple-choice lists; the student must formulate their own initial pathology hypothesis.



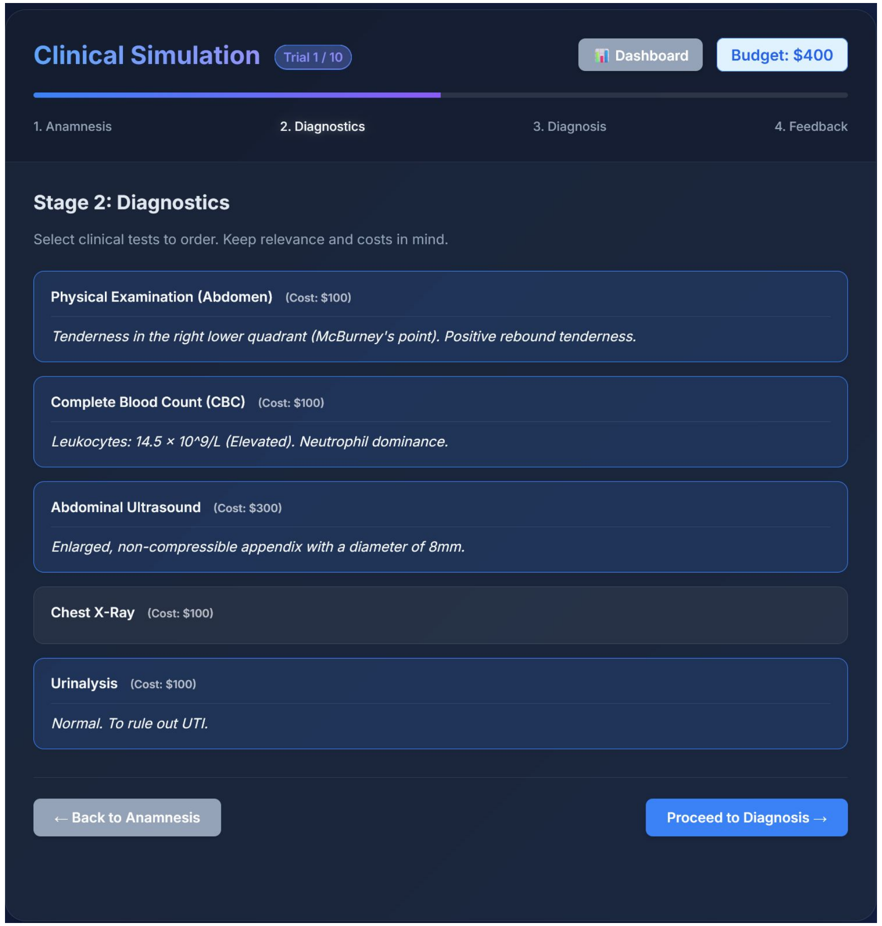
Active Exploration & Structural Scaffolding

- **Cognitive Apprenticeship (Collins et al., 1989):** Providing a situated, realistic environment (the ER) to develop cognitive skills.
- **Structural Scaffolding:** Strict 4-stage workflow (Anamnesis → Diagnostics → Diagnosis → Feedback) in the navigation bar.
- **Active Exploration:** Enforces history-gathering before diagnostics; prevents novices from rushing to expensive tests.
- **Didactic Design:** The student gathers subjective patient symptoms to refine their differential diagnosis list.



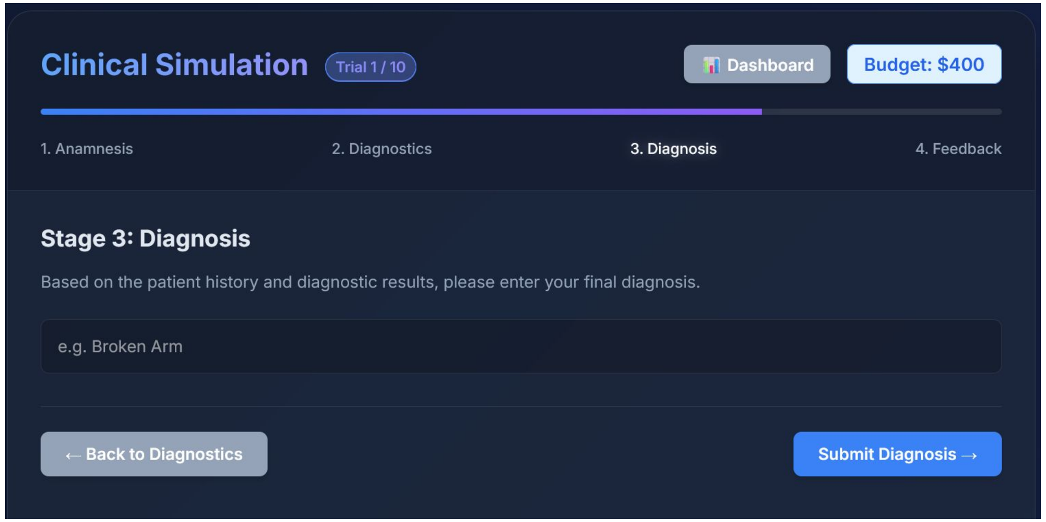
Managing Cognitive Load via Boundary Scaffolding

- **Boundary Scaffolding (Sweller, 1988):** Starting diagnostic budget manages cognitive load and sets explicit limits.
- **Combatting "Shotgun-Testing":** Novices tend to order every test available. The budget forces deliberate, hypothesis-driven clinical choices.
- **Active Experimentation (Kolb, 1984):** A penalty-free navigation loop. Students can move back and forth between Anamnesis ↔ Diagnostics to adjust their hypothesis.



Stimulating Articulation & Active Recall

- **Articulation (Collins et al., 1989):** Forcing students to externalize their implicit reasoning and diagnostic hypotheses.
- **Free-Text Entry vs. Recognition:** By using a text input instead of multiple-choice, we test active clinical recall rather than passive recognition.
- **Hypothesis Synthesis:** The student must synthesize anamnesis data and diagnostic findings to articulate a final, definitive medical diagnosis.
- **Clinical Accuracy Check:** The string is parsed and evaluated against target correct answers behind the scenes.

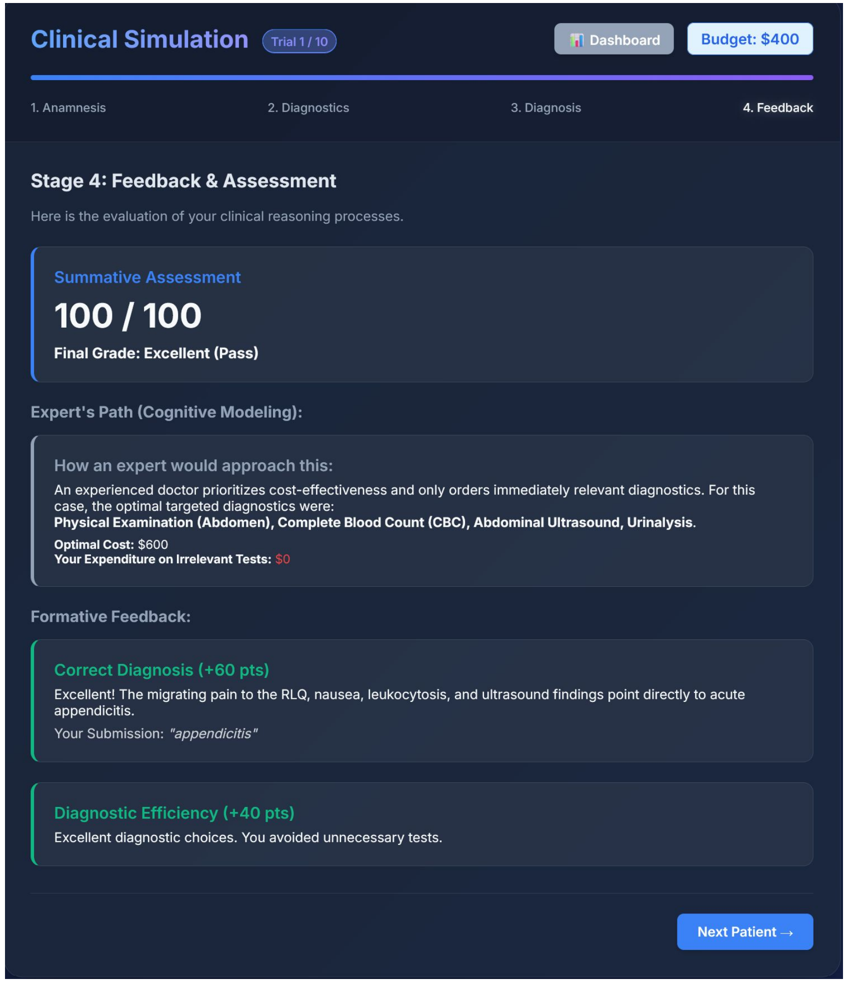


Clinical Evaluation: Assessment, Modeling & Feedback

1. Summative Assessment (Scriven, 1967): The final score (out of 100 points) and qualitative grade (e.g. "Excellent", "Good") evaluating overall performance.

2. Cognitive Modeling (Collins et al., 1989): A side-by-side comparison revealing exactly which diagnostics an expert physician would order and the optimal cost.

3. Formative Feedback (Black & Wiliam, 1998): Actionable clinical explanations clarifying both diagnostic correctness and diagnostic testing efficiency.



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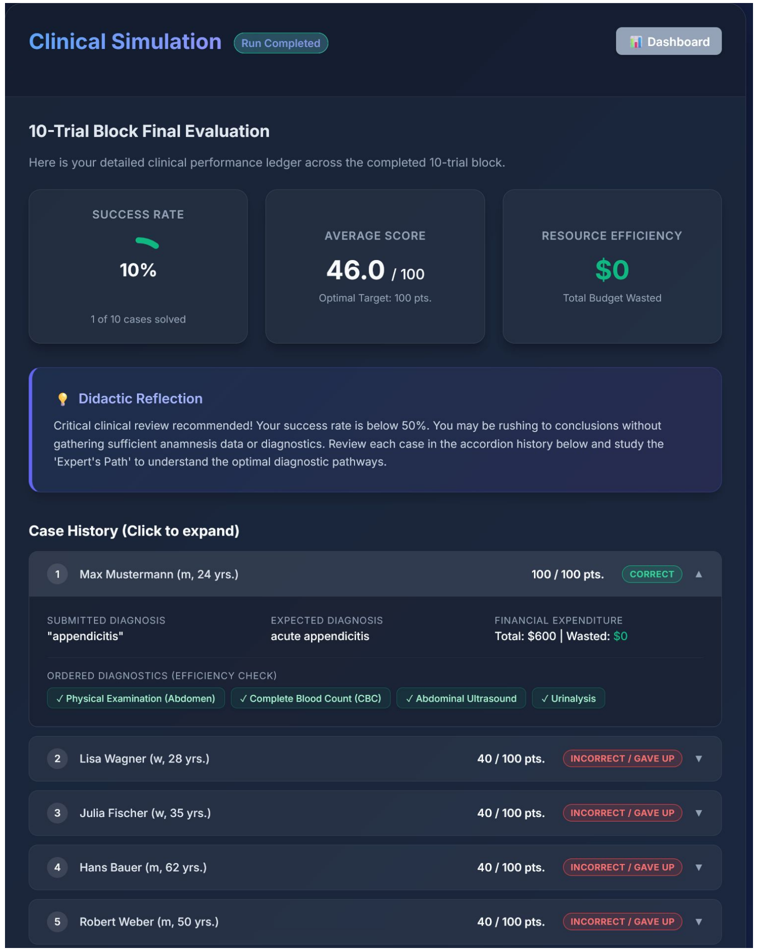
Metacognitive Reflection & Cognitive Auditing

Reflection-on-Action (Schön, 1983): Broadening focus from single-case outcomes to longitudinal cognitive patterns.

10-Trial Cohorts: Shuffles and ensures exposure to all 10 clinical conditions in a single block, with state saved across refreshes.

Didactic Coach: Analyzes metrics to give targeted metacognitive advice (e.g., warning accurate students about "shotgun-testing").

Accordion Ledger & Relevance Tags: Expandable ledger with color-coded tags ([✓ green] for relevant tests, [✗ red] for irrelevant tests) allows students to audit diagnostic search paths.



Conclusion: A Highly Cohesive Pedagogical Tool

Educational Concept	Technical Feature	Didactic Outcome
Problem-Based Learning	10 realistic, open-ended cases	Cultivates clinical hypothesis formulation
Cognitive Apprenticeship	Situated ER sandbox simulation	Makes expert diagnostic workflows explicit
Boundary Scaffolding	Strict 4-stage UI flow & Budget constraint	Prevents overload & shotgun-testing
Adaptive Fading	Budget decreases by $100 per case solved	Pushes from exploration to expert precision
Articulation & Recall	Free-text diagnosis input	Exercises active recall over passive recog.
Formative Feedback	Actionable clinical explanation & retry options	Enables iterative learning-from-errors
Summative Assessment	Final score (accuracy & efficiency check)	Enforces cost-effective clinical care
Learning Analytics	Persistent modal tracking longitudinal metrics	Stimulates self-regulated clinical learning
Metacognitive Reflection	10-Trial Block, coaching, tagged ledger	Promotes systematic reflection-on-action

THANK YOU FOR YOUR ATTENTION!